

**POKHARA UNIVERSITY**  
**GANDAKI COLLEGE OF ENGINEERING AND SCIENCE**

**LABORATORY REPORT**  
**AGILE SOFTWARE DEVELOPMENT LAB**

**Lab No. 1**  
BESE III YEAR - VI SEM

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Program: BESE ,2023

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## **NAME OF EXPERIMENT**

Managing project source code with Git commands and hosting the repository on GitHub.

## **AIM**

To gain practical understanding of Source Code Management (SCM) by using Git together with GitHub for recording changes, maintaining versions, and enabling collaborative software development.

## **THEORY**

Source Code Management (SCM), also called version control, refers to the discipline of recording and administering successive modifications made to program code. It gives a team the ability to work side by side, keep a complete history of every change, and roll back to an earlier state whenever something goes wrong.

Git is a distributed version control system. Because every developer keeps a full copy of the repository locally, work can continue offline and history is never dependent on a single central machine. Git records snapshots of the project as commits, each identified by a unique hash.

GitHub is a web-based service built around Git. In addition to storing remote repositories, it layers collaboration features on top of Git such as pull requests, code review, issue tracking, and project boards. Together, Git and GitHub form a backbone of Agile teams because they support frequent integration, transparent history, and continuous review of code.

## **OBJECTIVES**

- To become familiar with the core Git workflow.
- To create local repositories and manage them.
- To capture project progress through commits.
- To attach a local repository to a remote hosted on GitHub.
- To synchronise work by pushing and pulling changes.
- To appreciate how these tools support team-based development.

## **ALGORITHM**

1. Install Git and confirm the installation.
2. Register a free account on GitHub.
3. Create an empty remote repository on GitHub.
4. Initialise a Git repository inside the project folder.
5. Set the author name and email for commits.
6. Add the project files to the staging area.
7. Record the staged files as a commit.
8. Register the GitHub repository as the remote origin.
9. Push the committed work to GitHub.
10. Confirm that the local and remote repositories are in sync.

## **CODE**

### **Initialise the repository**

```
1 git init
```

### Set author identity

```
1 git config --global user.name "Pranish Poudel"
2 git config --global user.email "pranish@email.com"
```

### Inspect the working tree

```
1 git status
```

### Stage all files

```
1 git add .
```

### Record a commit

```
1 git commit -m "Initial commit"
```

### Attach the GitHub remote

```
1 git remote add origin https://github.com/username/repository.git
```

### Publish to GitHub

```
1 git branch -M main
2 git push -u origin main
```

### Fetch and merge remote work

```
1 git pull origin main
```

## OUTPUT

The Git repository was initialised and successfully connected to a remote hosted on GitHub.

- A new repository was created without errors.
- Project files were placed under Git tracking.
- The initial commit was recorded successfully.
- The remote origin was linked to the GitHub repository.
- The committed work was pushed to GitHub and appeared online.

## RESULT

Source Code Management using Git and GitHub was carried out successfully. Operations including repository creation, staging, committing, remote linkage, and pushing to GitHub were all performed and verified without issues.

## CONCLUSION

This lab gave hands-on familiarity with Source Code Management through Git and GitHub. It showed clearly how a version control system keeps a reliable record of code history, lets multiple contributors collaborate safely, and underpins the fast, iterative style of work that Agile Software Development depends on.